

Q: Derivating Normal equation from MLE

Source: PRML P141 ~

$$(\text{likelihood}) = P(\mathbf{t} | \mathbf{X}, \mathbf{W}, \beta) = \prod_{n=1}^N P(t_n | \mathbf{x}_n, \mathbf{W}, \beta)$$

$\therefore$ conditionally i.i.d given $\mathbf{X}, \mathbf{W}, \beta$

$$= \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{W}^T \phi(\mathbf{x}_n), \beta^{-1}) \quad \text{where} \quad \mathbf{W} = [\mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_{M-1}]^T$$
$$\phi(\mathbf{x}) = [\phi_0(\mathbf{x}), \phi_1(\mathbf{x}), \dots, \phi_{M-1}(\mathbf{x})]^T$$

takes log of likelihood,

$$\ln p(\mathbf{t} | \mathbf{X}, \mathbf{W}, \beta) = \ln \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{W}^T \phi(\mathbf{x}_n), \beta^{-1})$$

$$= \sum_{n=1}^N \ln \left(\left(\frac{\beta}{2\pi} \right)^{1/2} \cdot \exp \left[-\frac{\beta}{2} [t_n - \mathbf{W}^T \phi(\mathbf{x}_n)]^2 \right] \right)$$

$$= \sum_{n=1}^N \left(\frac{1}{2} \ln \beta - \frac{1}{2} \ln 2\pi - \frac{\beta}{2} [t_n - \mathbf{W}^T \phi(\mathbf{x}_n)]^2 \right)$$

$$= \frac{N}{2} \ln \beta - \frac{N}{2} \ln 2\pi - \underbrace{\beta \sum_{n=1}^N \frac{1}{2} [t_n - \mathbf{W}^T \phi(\mathbf{x}_n)]^2}_{\coloneqq \beta \cdot E_D(\mathbf{W})}$$

The gradient of log likelihood function is,

$$\nabla_w \log(\pi(x|w, \beta)) = -\nabla_w \beta \cdot E_D(w). \quad (\because \text{MLE is equivalent with least square error})$$

value of β does not change where makes this gradient zero. so we may dismiss β .

$$-\nabla_w E_D(w) = \frac{\partial}{\partial w} \left[\frac{1}{2} \sum_{n=1}^N (t_n - w^T \phi(x_n))^2 \right]$$

$$= -\frac{1}{2} \sum_{n=1}^N \frac{\partial}{\partial w} (t_n - w^T \phi(x_n))^2 = -\frac{1}{2} \sum_{n=1}^N 2 \cdot (t_n - w^T \phi(x_n)) (-\phi(x_n)^T)$$

($\because$ if a, x are vector, $\frac{\partial a^T x}{\partial a} = a$, we take transpose to $\phi(x_n)$ because gradient is row-vector form).

$$\therefore -\nabla_w E_D(w) = \sum_{n=1}^N (t_n - w^T \phi(x_n)) \phi(x_n)^T$$

Setting this gradient to zero gives

$$0 = \sum_{n=1}^N t_n \phi(x_n)^T - w^T \left(\sum_{n=1}^N \phi(x_n) \phi(x_n)^T \right)$$

$$\Leftrightarrow w^T \left(\sum_{n=1}^N \phi(x_n) \phi(x_n)^T \right) = \sum_{n=1}^N t_n \phi(x_n)^T$$

to make w^T to w , transposing both side of equal,

$$\left(\sum_{n=1}^N \phi(x_n) \phi(x_n)^T \right)^T w = \left(\sum_{n=1}^N t_n \phi(x_n)^T \right)^T$$

since $(A+B)^T = A^T + B^T$, $(AA^T)^T = AA^T$

$$\left(\sum_{n=1}^N \phi(x_n) \phi(x_n)^T \right) W = \sum_{n=1}^N t_n \phi(x_n)$$

to see intuition, unfolding sigma gives

$$\left(\phi(x_1) \phi(x_1)^T + \phi(x_2) \phi(x_2)^T \cdots \phi(x_n) \phi(x_n)^T \right) W = t_1 \begin{bmatrix} \phi_0(x_1) \\ \phi_1(x_1) \\ \vdots \\ \phi_n(x_1) \end{bmatrix} + t_2 \begin{bmatrix} \phi_0(x_2) \\ \phi_1(x_2) \\ \vdots \\ \phi_n(x_2) \end{bmatrix} + \cdots + t_n \begin{bmatrix} \phi_0(x_n) \\ \phi_1(x_n) \\ \vdots \\ \phi_n(x_n) \end{bmatrix}$$

defines $\Phi = [\phi(x_1) | \phi(x_2) | \cdots | \phi(x_n)]^T$

$$\Rightarrow \left(\phi(x_1) \phi(x_1)^T + \phi(x_2) \phi(x_2)^T \cdots \phi(x_n) \phi(x_n)^T \right) W = \Phi^T \cdot t$$

simplifies $\underbrace{\quad}_U$

$$U = \phi(x_1) \phi(x_1)^T + \phi(x_2) \phi(x_2)^T \cdots \phi(x_n) \phi(x_n)^T$$

$$= \underbrace{[\phi(x_1) | \phi(x_2) | \cdots | \phi(x_n)]}_{\Phi^T} \underbrace{\begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_n) \end{bmatrix}}_{\Phi} = \Phi^T \Phi$$

$$\therefore \Phi^T \Phi W = \Phi^T t$$

if $\Phi^T \Phi$ is invertible,

$$W = (\Phi^T \Phi)^{-1} \Phi^T t$$

(re-derivate with column form gradient)

$$-\nabla_w (E_D(W)) = \sum_{n=1}^N (t_n - W^T \phi(x_n)) \phi(x_n)$$

$$-\nabla_w E_D(W) = 0 \Rightarrow \sum_{n=1}^N (W^T \phi(x_n)) \phi(x_n) = \sum_{n=1}^N t_n \phi(x_n)$$

∴ note that Associative property doesn't hold because $W^T(\phi(x_n) \phi(x_n))$ doesn't exist.

$$\Rightarrow \left(\sum_{n=1}^N (W^T \phi(x_n)) \phi(x_n) \right)^T = \left(\sum_{n=1}^N t_n \phi(x_n) \right)^T$$

$$\Rightarrow \sum_{n=1}^N \phi(x_n)^T (\phi(x_n)^T W) = \sum_{n=1}^N t_n \phi(x_n)^T$$

$$\begin{aligned} \sum_{n=1}^N \phi(x_n)^T (\phi(x_n)^T W) &= t^T \begin{bmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix} \\ &= t^T \underbrace{\begin{bmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix}}_{:= \Phi} \end{aligned}$$

$$\left(\sum_{n=1}^N \phi(x_n)^T (\phi(x_n)^T w) \right) = (t^T \Phi)$$

$$\Rightarrow \left(\sum_{n=1}^N \phi(x_n)^T (\phi(x_n)^T w) \right) = (t^T \Phi)$$

$$\Rightarrow \sum_{n=1}^N (\phi(x_n)^T w) \phi(x_n)^T = t^T \Phi$$

$$\Rightarrow [\phi(x_1)^T w \mid \phi(x_2)^T w \mid \dots \mid \phi(x_n)^T w] \begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_n) \end{bmatrix} = t^T \Phi$$

$\underbrace{\hspace{10em}}_{\Phi}$

$$\Rightarrow [\phi(x_1)^T w \mid \phi(x_2)^T w \mid \dots \mid \phi(x_n)^T w] \Phi = t^T \Phi$$

$$\frac{[\quad] [w] \mid [\quad] [w] \rightarrow [w] [\quad] \mid [w] [\quad]}{[\quad] [w] \mid [\quad] [w] \rightarrow [w] [\quad] \mid [w] [\quad]}$$

$$\Rightarrow [w^T \phi(x_1) \mid w^T \phi(x_2) \mid \dots \mid w^T \phi(x_n)]^T \quad (\text{since scalar is symmetric})$$

$$\Phi = t^T \Phi$$

$$\Rightarrow \underbrace{w^T [\phi(x_1) \mid \phi(x_2) \mid \dots \mid \phi(x_n)]}_{\Phi^T} \Phi = t^T \Phi$$

=)

$$\Rightarrow W^T \Phi^T \Phi = t^T \Phi$$

$$\Rightarrow (W^T \Phi^T \Phi)^T = (t^T \Phi)^T$$

$$\Rightarrow \Phi^T \Phi W = \Phi^T t$$
